Chain-of-Thought's Double Edge Is Model-Specific: Prompting versus Scale Across Three Model Families

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Abstract.

Getting more out of a language model comes down to two solutions: either invest in a better model, or create a better prompt for the current one. This paper quantifies that choice with a **parameter-equivalent exchange rate**, a measure of how much model scaling a prompting strategy is worth. Holding 4-bit quantization constant, three instruction-tuned families were tested: Qwen2.5 (0.5B to 7B), Llama-3 (1B to 8B), and Gemma-2 (2B and 9B), across four strategies (zero-shot, few-shot, chain-of-thought, and structured JSON) on sentiment (SST-2), knowledge (MMLU), and multi-step arithmetic (GSM8K). **On reasoning the exchange rate is large: a good prompt is worth close to a 5× increase in parameters**, with a 1.5B Qwen using CoT (58.7%) outscoring a 7B Qwen without it (15.3%). This comparison holds in Llama (3B CoT 73.0% vs 8B zero-shot 20.1%) and Gemma (2B CoT 54.0% vs 9B zero-shot 16.3%). On knowledge there is little to no change with prompting, and accuracy tracks parameter count. This split sharpens the meta-analysis of Sprague et al. (2024). The knowledge result then *departs* from Sprague et al., who report CoT as roughly neutral on knowledge tasks on average. That average hides a heterogeneous, family-specific pattern: in Qwen, CoT lowers MMLU by 7 to 16 points, with the loss growing as the model scales; in Gemma-2 it hurts the 2B model (−16pp) but fades to neutral by 9B (+0.6pp); in Llama it stays small with no trend (+0.9 to −5.2pp). Pooling these very different responses into one "no effect on average" describes none of them and masks a real, structured, per-family effect. A continuous numeric-presence metric and re-extraction from full, untruncated outputs prove these effects are not scoring artifacts. An unquantized (bf16) control on the Qwen ladder reproduces the CoT penalty, larger if anything, so quantization does not create it. Every result was created on a single 16GB laptop, so the multiple-tier reasoning gain a good prompt buys is within reach of an ordinary user, without a larger model or specialized hardware. Code and prompts are at github.com/ssamalsamir/cot-model-specificity.

1. Introduction

Running a language model in practice means picking a model and picking how to prompt it, and the two compete for the same budget. A bigger model costs compute, memory, and latency. A better prompt is close to free, but its benefit is bounded. This asymmetry has a practical upside: whenever a prompt can stand in for a much larger model, that capability becomes available to anyone with a commodity laptop, not only to those who can afford to run or serve a bigger one. Every result in this paper was produced on a single 16GB consumer machine, which makes the question of *how much* a prompt is worth a question about who can access strong reasoning at all. The two levers have mostly been studied apart: few-shot ability grows with scale (Brown et al., 2020), chain-of-thought helps larger models reason (Wei et al., 2022b; Kojima et al., 2022), and accuracy follows smooth scaling laws (Kaplan et al., 2020; Hoffmann et al., 2022). What is missing is a direct measurement of how they trade off on one controlled model ladder.

Two questions drive the study. **RQ1 (exchange rate):** how much model scaling does each prompting strategy substitute for, and does the answer depend on the task? **RQ2 (CoT's cost):** does a strategy that helps reasoning carry over to other tasks, or can it hurt?

Sprague et al. (2024) report, across many models, that chain-of-thought helps mainly on math and symbolic reasoning and adds little on knowledge tasks. This paper reproduces that split on a controlled, same-family, sub-7B ladder and puts a number on it: a good prompt is worth close to a 5× increase in parameters on GSM8K, and almost nothing on MMLU, where the facts either sit in the weights or they do not. The result then goes further than a reproduction. The conclusion that CoT is roughly neutral on knowledge is a population *average*, and that average hides a heterogeneous, family-specific pattern: across the three families, CoT on knowledge questions ranges from clearly harmful and worsening with scale (Qwen), to harmful only at small scale (Gemma-2), to roughly flat (Llama-3). A single average over models therefore reports "no effect" where a real, structured, per-family effect is present (§4.3).

Contributions. (1) The prompt–parameter exchange rate is quantified and shown to be large on reasoning: a good prompt is worth close to a 5× increase in parameters on GSM8K, and near zero on knowledge. (2) Chain-of-thought's effect on knowledge QA differs by family across the three families. CoT lowers Qwen2.5's MMLU accuracy by 7 to 16 points, with the loss growing as the model gets larger; it hurts Gemma-2 at 2B but not at 9B; and it stays small with no scale trend in Llama-3. This qualifies the meta-analysis of Sprague et al. (2024): a near-zero average effect of CoT on knowledge is an artifact of averaging over families with different, size-dependent responses, not evidence that CoT does nothing. The effect survives checks for extraction, metric, and quantization artifacts; an unquantized bf16 control on the Qwen ladder reproduces the MMLU penalty and in fact enlarges it. (3) The analysis is more careful than is typical at this scale: three model families, a single fixed evaluation set of up to 1000 items per condition, Wilson confidence intervals, and Holm-adjusted McNemar tests. (4) The whole harness runs on one consumer laptop, and the code and prompts are released.

2. Related Work

Prompting and scale. Few-shot in-context learning improves with model size (Brown et al., 2020), and chain-of-thought prompting mainly helps larger models reason (Wei et al., 2022b; Kojima et al., 2022). The interaction between few-shot benefit and model size has been studied directly (San Joaquin & Haroen, 2022), and small models are known to struggle with long reasoning chains (Li et al., 2025). Closest to the present setting, Sprague et al. (2024) show across many models and datasets that CoT's gains concentrate in math and symbolic reasoning and are small on knowledge and language-understanding tasks. The present study reproduces that split on a controlled same-family, sub-7B ladder and adds a finding they do not test for: the size-dependent response can differ sharply between families. None of this prior work reports a prompt-as-parameters exchange rate on a controlled same-family ladder, or the family-specific knowledge-QA loss reported here.

Scaling and emergence. Performance follows power-law scaling (Kaplan et al., 2020; Hoffmann et al., 2022). Some abilities seem to appear abruptly with scale (Wei et al., 2022a), although that "emergence" may be partly an artifact of discontinuous metrics (Schaeffer et al., 2023). That artifact is checked in §4.4.

Format and prompt sensitivity. Required output formats can degrade reasoning (Tam et al., 2024). Meaning-preserving format changes move accuracy a lot, and the sensitivity does not wash out with scale (Sclar et al., 2023). Few-shot example ordering alone can dominate the result (Lu et al., 2022), and GSM8K may reward brittle pattern-matching rather than reasoning (Mirzadeh et al., 2024). The nearest comparisons are Tam et al. (2024), who vary format restrictions but across a mixed set of models rather than a controlled size ladder, and San Joaquin & Haroen (2022). Neither frames the result as an exchange rate, and neither contrasts knowledge against multi-step reasoning in this way.

3. Methods

Models. Three instruction-tuned families are used, with 4-bit quantization held fixed, so size is the only thing that varies within a family: Qwen2.5-Instruct at 0.5B / 1.5B / 3B / 7B, Llama-3 at 1B / 3B / 8B, and Gemma-2 at 2B / 9B. Everything runs locally through MLX on an Apple M4 (16GB), with one model loaded at a time.

Tasks. SST-2 (binary sentiment), MMLU (4-way multiple-choice knowledge, stratified across all 57 subjects), and GSM8K (grade-school multi-step arithmetic). Each condition is scored on the same fixed evaluation set: the full SST-2 validation set (872 items), and a single fixed sample of 1000 items each for GSM8K and MMLU, held constant across every model and strategy. Decoding is greedy and therefore deterministic, so the only thing that varies between conditions is the model and the prompt, not the choice of examples. A single large fixed set gives more paired statistical power than splitting the data into smaller repeated draws.

Strategies. Zero-shot; few-shot (fixed hand-written exemplars); CoT (reasoning followed by a parsed final answer); structured (JSON-only output). Full templates are in Appendix A.

Decoding and scoring. Greedy decoding (temperature 0), so runs are deterministic. Answers are extracted per task: numeric match for GSM8K, exact label or letter match otherwise.

Statistics. Accuracy is reported with 95% Wilson score intervals (about ±3pp at $n = 1000$) and paired McNemar tests (continuity-corrected, Holm-adjusted) for the within-model strategy comparisons. The **exchange rate** is defined plainly: prompting at size S "matches" scaling if the best-prompt accuracy at S is at least the zero-shot accuracy at the next size up.

Robustness. Exact-match scoring can exaggerate gaps between small and large models (Schaeffer et al., 2023). As a check on GSM8K, the harness also records *numeric-present* (whether the correct value appears anywhere in the output) and the relative error of the committed answer, and re-extracts answers from the full, untruncated outputs (§4.4).

4. Results

4.1 Main results

Scaling curves by strategy

Figure 1. Accuracy vs. model scale (log axis) for each prompting strategy, by family (rows) and task (columns). On GSM8K, the CoT curve (red) separates sharply from all others; on MMLU the strategies bunch together and rise mainly with scale; on SST-2 the spread is moderate. Note Llama-3's erratic few-shot curve (blue).

Task	Model	zero-shot	few-shot	CoT	structured
SST-2	Qwen-0.5B	61.0	**87.6**	67.4	73.1
	Qwen-1.5B	85.1	**92.0**	88.8	91.4
	Qwen-3B	80.4	82.3	89.3	**90.9**
	Qwen-7B	83.7	90.4	**93.5**	92.7
MMLU	Qwen-0.5B	**35.2**	27.5	28.0	26.1
	Qwen-1.5B	55.6	54.7	45.3	**56.9**
	Qwen-3B	**60.7**	59.6	47.0	60.7
	Qwen-7B	70.2	**71.7**	54.7	69.9
GSM8K	Qwen-0.5B	2.7	3.2	**24.7**	2.0
	Qwen-1.5B	3.4	7.2	**58.7**	12.0
	Qwen-3B	1.7	1.7	**71.1**	2.0
	Qwen-7B	15.3	2.0	**88.7**	22.4

Table 1. Qwen2.5 accuracy (%). Bold = best strategy in that row; red = CoT below zero-shot.

Task	Model	zero-shot	few-shot	CoT	structured
SST-2	Llama-1B	58.4	0.1	70.6	**73.4**
	Llama-3B	88.6	55.7	84.7	**89.9**
	Llama-8B	**92.9**	85.4	90.9	92.3
MMLU	Llama-1B	32.4	1.3	33.3	**39.0**
	Llama-3B	52.9	39.7	55.9	**56.6**
	Llama-8B	62.5	56.3	57.3	**63.6**
GSM8K	Llama-1B	**6.6**	1.4	1.4	4.8
	Llama-3B	11.1	1.5	**73.0**	7.7
	Llama-8B	20.1	0.8	**81.9**	9.9

Table 2. Llama-3 accuracy (%). Bold = best in row; red marks a strategy below zero-shot (direction only; not every gap is significant). Few-shot falls to near zero on SST-2 and MMLU, a signature consistent with an exemplar-formatting failure rather than a graded capability result. The key cross-family difference: Llama's CoT shows none of Qwen's large, scale-increasing MMLU penalty, reaching at worst –5.2pp (8B) against Qwen's –15.5pp.

Task	Model	zero-shot	few-shot	CoT	structured
SST-2	Gemma-2B	89.9	74.3	**90.5**	88.6
	Gemma-9B	90.6	89.8	90.6	**92.2**
MMLU	Gemma-2B	55.2	52.1	39.1	**56.5**
	Gemma-9B	69.7	69.1	**70.3**	68.3
GSM8K	Gemma-2B	5.7	2.5	**54.0**	5.4
	Gemma-9B	16.3	2.7	**86.3**	21.8

Table 3. Gemma-2 accuracy (%). Bold = best in row; red = strategy below zero-shot. The MMLU CoT penalty is large at 2B (−16.1pp) but gone by 9B (+0.6pp), a "harms-then-fades" shape across the two available checkpoints, distinct from Qwen (penalty grows with scale) and Llama (flat). GSM8K shows the same CoT exchange rate as the other families (Gemma-2B CoT 54.0 vs Gemma-9B zero-shot 16.3).

4.2 The exchange rate is large on reasoning and small on knowledge (RQ1)

Exchange rate on GSM8K

Figure 3. Zero-shot vs. CoT accuracy on GSM8K. In all three families, CoT at a small size sits far above zero-shot at a much larger size. For example, Qwen-1.5B + CoT (58.7%) exceeds Qwen-7B zero-shot (15.3%); Llama-3B + CoT (73.0%) exceeds Llama-8B zero-shot (20.1%); and Gemma-2B + CoT (54.0%) exceeds Gemma-9B zero-shot (16.3%).

GSM8K: prompting beats scaling. CoT at every size exceeds zero-shot at the next size up. A 1.5B Qwen with CoT beats a 7B Qwen without it by 43pp, so on this task the prompt is worth roughly a 5× increase in parameters, and the same holds in Llama and Gemma (Gemma-2B CoT 54.0 vs Gemma-9B zero-shot 16.3). The gain is specific to CoT: few-shot and structured stay near the floor, because answer-only or format-constrained prompts cut off the reasoning the task needs. Llama-1B is the exception, too weak to reason even with CoT.

MMLU: only scale helps. The best strategy beats zero-shot by at most a few points (≤1.5pp in Qwen, ≤6.6pp in Llama, ≤1.3pp in Gemma). Accuracy climbs with parameters no matter the prompt. A fact is either in the weights or it is not. This split between reasoning and knowledge matches, on a controlled small-model ladder, the broad pattern reported by Sprague et al. (2024).

SST-2: prompting helps a moderate amount, with best-strategy gains up to about 27pp, often around one model tier's worth.

4.3 CoT helps reasoning but can hurt knowledge, and that depends on the family (RQ2)

CoT minus zero-shot by task and family

Figure 2. CoT minus zero-shot accuracy (pp) vs. model size. On GSM8K all three families are large and positive. On MMLU they diverge: Qwen (purple) drops increasingly below zero (worst at 7B: −15.5pp); Gemma (blue) is sharply negative at 2B (−16.1pp) but recovers to zero by 9B; Llama (orange) stays small with no trend (+0.9 to −5.2pp). On SST-2 all are small.

In Qwen2.5, the CoT prompt that wins on GSM8K lowers MMLU accuracy at every size: by 7.2pp (0.5B), 10.3pp (1.5B), 13.7pp (3B), and 15.5pp (7B), and the loss grows as the model gets larger. One plausible mechanism, consistent with the data but not established by it, is over-reasoning on multiple-choice items, where free-form reasoning gives the model room to talk itself out of the correct option (cf. Tam et al., 2024). The zero-shot and CoT prompts also differ in wording, answer format, and token budget, so the contrast is not a pure reasoning ablation. On average across many models, Sprague et al. (2024) find CoT to be roughly neutral on MMLU, so an active loss that grows with scale inside a single family is a sharper effect than has been reported before.

The other two families behave differently. Gemma-2 is a third case: CoT hurts MMLU sharply at 2B (−16.1pp) but the penalty is gone by 9B (+0.6pp), so here the harm *fades* with scale rather than growing as in Qwen, though with only two checkpoints this is a suggestive shape rather than an established trend. Llama-3 is smaller still and shows no scale trend (+0.9pp at 1B, +3.0pp at 3B, −5.2pp at 8B), and its 1B and 3B differences are not statistically significant (§C.2), which reinforces the flat reading. The three families thus give three different CoT-on-knowledge responses: growing harm (Qwen), harm-then-recovery (Gemma), and roughly flat (Llama). CoT's harm to knowledge QA is therefore family-dependent rather than a general rule. Pooling these unlike responses into one average is misleading: it hides the large, scale-worsening loss in Qwen behind the milder Gemma and Llama numbers, producing a single net figure that describes none of the three. That averaging is exactly what produces a verdict of "CoT is roughly neutral on knowledge" when results are combined across heterogeneous models, as in the meta-analysis of Sprague et al. (2024). Resolved per family, the neutral-looking average is revealed as a blend of unlike effects, not an absence of effect, so the present result qualifies that conclusion rather than merely extending it. The broader claim from §4.2, that prompting buys little on knowledge whatever the strategy, still holds in all three families; what does not hold is that CoT is harmless there. A study of any one family alone would have drawn the wrong general conclusion.

Structured (JSON) output is not a blanket penalty in any family. It is among the strongest strategies on SST-2 (Qwen-7B: 92.7; Llama-8B: 92.3), competitive on MMLU, and clearly harmful only on GSM8K, which narrows the "format tax" of Tam et al. (2024) to a reasoning-specific cost. Few-shot, on the other hand, is much less reliable in Llama-3 than in Qwen2.5 (for example GSM8K around 1%, and SST-2 at 1B just 0.1%), in line with the known family-specific sensitivity to how exemplars are formatted (Sclar et al., 2023; Lu et al., 2022).

4.4 The effects are not scoring artifacts

Continuous metric (GSM8K). Numeric-present stays close to exact-match instead of pulling apart from it (Qwen-0.5B CoT 24.7 / 38.2; Qwen-7B CoT 88.7 / 91.5), and the median relative error of the committed answer is 0.000 for CoT at 1.5B and above. So the low small-model scores are not an artifact of a discontinuous metric (Schaeffer et al., 2023).

Extraction audit (MMLU CoT). Re-extracting answers from the full outputs, the extraction-failure rate is only 2 to 4%. Scoring just the cleanly-parsed answers, Qwen CoT still trails zero-shot by 7 to 16pp at every size. The CoT-hurts-knowledge effect in Qwen is real, not a parsing failure.

Quantization control (MMLU, Qwen). Because the main results hold 4-bit quantization fixed, the Qwen CoT-on-knowledge penalty could in principle be a quantization artifact. As a test, the Qwen ladder (0.5B / 1.5B / 3B) was re-run at full precision (bf16) on the same fixed MMLU set. The penalty does not merely survive, it grows: CoT trails zero-shot by 12.9 / 14.2 / 20.9pp at bf16, against 7.2 / 10.3 / 13.7pp at 4-bit, and still increases monotonically with scale. Quantization was therefore *masking* part of the effect rather than creating it, so the Qwen MMLU penalty is not a low-precision artifact. Per-condition bf16 accuracies, sample sizes, and tests are listed in Appendix C.3.

5. Discussion

The central observation is that one prompting technique can affect three families three different ways on the same task. Chain-of-thought lifts all three on math, but on multiple-choice knowledge it lowers Qwen's accuracy more and more with scale, hurts Gemma-2 at 2B but not at 9B, and leaves Llama roughly where it started. Within Qwen the penalty grows at every step up the size ladder, which makes it unlikely to be noise or a one-checkpoint fluke; Gemma shows the opposite scale relationship, and Llama shows almost none. Three families make this a more robust observation than a two-family contrast, but "family-specific" is best treated as something to test more widely, not as a settled rule. A plausible mechanism is over-reasoning: thinking out loud gives a model room to talk itself out of a correct answer, and the families may differ in how prone they are to that.

The broader prompting-versus-scale pattern has a simple reading. Reasoning tasks hold latent ability that a prompt can draw out, whereas a fact is either stored in the weights or it is not, so only scale moves it (Sprague et al., 2024). The practical takeaway: under a fixed budget, spend on prompting for reasoning workloads and on parameters for knowledge workloads, and do not assume chain-of-thought is safe on knowledge questions, because whether it helps depends on the model. The reasoning half of that rule has a wider consequence. Because the prompting lever is nearly free and runs on commodity hardware, the measured exchange rate lowers the bar for who can obtain strong reasoning at all: a laptop-sized model with the right prompt can stand in for one several times larger, so the gap between users who can afford big models and those who cannot is, on reasoning tasks, partly closable with a prompt.

6. Limitations

- **Three model families.** The CoT-on-knowledge effect is consistent across every size within each of Qwen, Gemma, and Llama, but three families still cannot establish it as a general property; more families (e.g., Phi, Mistral) and additional checkpoints are needed to confirm it generalizes.
- **4-bit quantization throughout.** Held constant across the main results. For the headline Qwen MMLU CoT penalty, an unquantized (bf16) control (§4.4) reproduces the effect, and in fact enlarges it, so quantization does not explain it. The full cross-family grid (Llama, Gemma, and the GSM8K and SST-2 tasks) was not re-run at full precision, so a residual quantization interaction elsewhere cannot be entirely ruled out.
- **One evaluation set per condition.** Each condition is scored on a single fixed sample of 872 to 1000 items. Because decoding is greedy, there is no run-to-run variance to report; the Wilson intervals (about ±3pp) bound the sampling error. Accuracies within a few points of each other should be read as ties, not as differences.
- **Greedy decoding only.** No self-consistency and no temperature sweep, either of which could change the CoT results.
- **Fixed hand-written exemplars.** Prompt order and exemplar format were not ablated (Lu et al., 2022; Sclar et al., 2023); Llama's near-zero few-shot scores are most likely an instance of this formatting sensitivity rather than a capability result.
- **Scope.** Three benchmarks, English only. GSM8K may test brittle pattern-matching more than reasoning (Mirzadeh et al., 2024), so the "reasoning" label comes with that caveat.
- **Metric.** The numeric-present metric addresses the Schaeffer (2023) concern, but exact-match remains the headline number and both are reported. The exchange rate is a descriptive measure, not a law.

7. Conclusion

The central, quantitative result is the exchange rate: on reasoning, a good prompt is worth close to a 5× increase in parameters, since a 1.5B model with chain-of-thought beats a 7B model without it, while on knowledge it buys almost nothing. That single number reframes prompt engineering from a soft "it helps" into a concrete substitute for scale, and because it holds on a 16GB laptop, it puts large-model reasoning within reach of anyone, not only those who can run a bigger model.

On knowledge, the prevailing view is qualified rather than merely extended. Chain-of-thought's effect there depends on the model: the penalty grows with scale in Qwen2.5, appears at 2B but fades by 9B in Gemma-2, and stays small throughout in Llama-3, giving three families three responses on the same task. Pooled across models, these unlike effects blend into the "roughly neutral" average reported by Sprague et al. (2024); resolved per family, that average is revealed as a mix of real effects, not their absence. The practical lesson is that which lever to pull depends on both the task and the model, and that a prompt which helps one model cannot be assumed safe on another.

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Appendix A. Prompt templates

Each prompt below is wrapped in the model's chat template (`apply_chat_template`, `add_generation_prompt=True`) before generation. `{...}` denotes the inserted example.

A.1 SST-2 (sentiment)

```
zero-shot:  Classify the sentiment of the following sentence as exactly
            'positive' or 'negative'.
```

```
Sentence: {sentence}
Sentiment:

few-shot:   Classify the sentiment as 'positive' or 'negative'.
            [3 exemplars: ("The film is a masterpiece of visual storytelling." -> posi-
tive),
             ("I fell asleep twice. Complete waste of time." -> negative),
             ("A quietly moving portrait of grief and resilience." -> positive)]

Sentence: {sentence}
Sentiment:

CoT:        Identify the key sentiment signals in the sentence, reason through them,
            then classify as 'positive' or 'negative'. End your response with
            'Final answer: positive' or 'Final answer: negative'.
Sentence: {sentence}
Reasoning:

structured: Classify sentiment. Respond ONLY with valid JSON:
            {"sentiment": "positive"} or {"sentiment": "negative"}.
Sentence: {sentence}
JSON:
```

A.2 GSM8K (arithmetic)

```
zero-shot:  Solve the following math problem. Give only the final number as your
            answer.

Problem: {question}
Answer:

few-shot:   2 worked examples shown as Problem / Answer (final number only),
            then Problem: {question}
Answer:

CoT:        2 worked examples shown as Problem / Reasoning / Answer, then
            "Solve ... step by step. End with 'Answer: <number>'."

Problem: {question}
Reasoning:

structured: Solve this math problem. Respond ONLY with valid JSON:
            {"answer": "<number>"}.
Problem: {question}
JSON:
```

A.3 MMLU (knowledge, 4-way MC)

```
zero-shot:  Answer the following multiple choice question. Respond with only the
            letter A, B, C, or D.

Question: {q}
{A..D options}
Answer:
```

```
few-shot:   4 exemplars (answers spread across A/B/C/D to avoid position bias),
            then Question: {q}
{options}
Answer:

CoT:        "Think through each option carefully, then end your response with
            'Final answer: A' (or B, C, D)."
Question: {q}
{options}
Reasoning:

structured: Respond ONLY with valid JSON: {"answer": "A"} (or B, C, D).

Question: {q}
{options}
JSON:
```

Appendix B. Answer-extraction rules

Greedy decoding (temperature 0); answers parsed from raw model text as follows.

- **SST-2.** structured: parse JSON `sentiment` field. CoT: regex `final answer[:\s]+(positive|negative)`. Otherwise: scan the last 60 characters, then the whole output, for "positive"/"negative".
- **GSM8K.** structured: parse JSON `answer`. Otherwise: first match of `answer[:\s]+$?(number)`, then a trailing `= $number`, then the last number in the text (excluding bare years 2020-2026). Scored by numeric equality (tolerance 0.01).
- **MMLU.** structured: parse JSON `answer`. CoT: regex `final answer[:\s]+([A-D])`. Otherwise: a lone A-D letter on the last non-empty line, then the last A-D letter anywhere. Scored by exact letter match.

Extraction failures count as incorrect. An audit of MMLU-CoT on full (untruncated) outputs found a 2-4% extraction-failure rate, with the reported effects unchanged (§4.4).

Appendix C. Full per-condition results (95% Wilson CIs & McNemar tests)

Each condition is scored on the same fixed evaluation set: the full SST-2 validation set (872 items) and a fixed sample of 1000 items each for GSM8K and MMLU (see the n column), held constant across every model and strategy. McNemar tests are continuity-corrected and Holm-adjusted within each model×task, comparing each strategy to zero-shot on the same items.

C.1 Accuracy and 95% Wilson CI

Family	Model	Task	Strategy	n	Acc (%)	95% CI
Qwen2.5-Instruct	qwen-0.5b	sst2	zero-shot	872	61.0	[57.7, 64.2]
Qwen2.5-Instruct	qwen-0.5b	sst2	few-shot	872	87.6	[85.3, 89.6]
Qwen2.5-Instruct	qwen-0.5b	sst2	CoT	872	67.4	[64.3, 70.5]
Qwen2.5-Instruct	qwen-0.5b	sst2	structured	872	73.1	[70.0, 75.9]
Qwen2.5-Instruct	qwen-0.5b	gsm8k	zero-shot	1000	2.7	[1.9, 3.9]
Qwen2.5-Instruct	qwen-0.5b	gsm8k	few-shot	1000	3.2	[2.3, 4.5]
Qwen2.5-Instruct	qwen-0.5b	gsm8k	CoT	1000	24.7	[22.1, 27.5]
Qwen2.5-Instruct	qwen-0.5b	gsm8k	structured	1000	2.0	[1.3, 3.1]
Qwen2.5-Instruct	qwen-0.5b	mmlu	zero-shot	1000	35.2	[32.3, 38.2]
Qwen2.5-Instruct	qwen-0.5b	mmlu	few-shot	1000	27.5	[24.8, 30.3]
Qwen2.5-Instruct	qwen-0.5b	mmlu	CoT	1000	28.0	[25.3, 30.9]
Qwen2.5-Instruct	qwen-0.5b	mmlu	structured	1000	26.1	[23.5, 28.9]
Qwen2.5-Instruct	qwen-1.5b	sst2	zero-shot	872	85.1	[82.6, 87.3]
Qwen2.5-Instruct	qwen-1.5b	sst2	few-shot	872	92.0	[90.0, 93.6]
Qwen2.5-Instruct	qwen-1.5b	sst2	CoT	872	88.8	[86.5, 90.7]
Qwen2.5-Instruct	qwen-1.5b	sst2	structured	872	91.4	[89.4, 93.1]
Qwen2.5-Instruct	qwen-1.5b	gsm8k	zero-shot	1000	3.4	[2.4, 4.7]
Qwen2.5-Instruct	qwen-1.5b	gsm8k	few-shot	1000	7.2	[5.8, 9.0]
Qwen2.5-Instruct	qwen-1.5b	gsm8k	CoT	1000	58.7	[55.6, 61.7]
Qwen2.5-Instruct	qwen-1.5b	gsm8k	structured	1000	12.0	[10.1, 14.2]
Qwen2.5-Instruct	qwen-1.5b	mmlu	zero-shot	1000	55.6	[52.5, 58.7]
Qwen2.5-Instruct	qwen-1.5b	mmlu	few-shot	1000	54.7	[51.6, 57.8]
Qwen2.5-Instruct	qwen-1.5b	mmlu	CoT	1000	45.3	[42.2, 48.4]
Qwen2.5-Instruct	qwen-1.5b	mmlu	structured	1000	56.9	[53.8, 59.9]
Qwen2.5-Instruct	qwen-3b	sst2	zero-shot	872	80.4	[77.6, 82.9]
Qwen2.5-Instruct	qwen-3b	sst2	few-shot	872	82.3	[79.7, 84.7]
Qwen2.5-Instruct	qwen-3b	sst2	CoT	872	89.3	[87.1, 91.2]
Qwen2.5-Instruct	qwen-3b	sst2	structured	872	90.9	[88.9, 92.7]
Qwen2.5-Instruct	qwen-3b	gsm8k	zero-shot	1000	1.7	[1.1, 2.7]
Qwen2.5-Instruct	qwen-3b	gsm8k	few-shot	1000	1.7	[1.1, 2.7]
Qwen2.5-Instruct	qwen-3b	gsm8k	CoT	1000	71.1	[68.2, 73.8]
Qwen2.5-Instruct	qwen-3b	gsm8k	structured	1000	2.0	[1.3, 3.1]
Qwen2.5-Instruct	qwen-3b	mmlu	zero-shot	1000	60.7	[57.6, 63.7]
Qwen2.5-Instruct	qwen-3b	mmlu	few-shot	1000	59.6	[56.5, 62.6]
Qwen2.5-Instruct	qwen-3b	mmlu	CoT	1000	47.0	[43.9, 50.1]

Qwen2.5-Instruct	qwen-3b	mmlu	structured	1000	60.7	[57.6, 63.7]
Qwen2.5-Instruct	qwen-7b	sst2	zero-shot	872	83.7	[81.1, 86.0]
Qwen2.5-Instruct	qwen-7b	sst2	few-shot	872	90.4	[88.2, 92.2]
Qwen2.5-Instruct	qwen-7b	sst2	CoT	872	93.5	[91.6, 94.9]
Qwen2.5-Instruct	qwen-7b	sst2	structured	872	92.7	[90.7, 94.2]
Qwen2.5-Instruct	qwen-7b	gsm8k	zero-shot	1000	15.3	[13.2, 17.7]
Qwen2.5-Instruct	qwen-7b	gsm8k	few-shot	1000	2.0	[1.3, 3.1]
Qwen2.5-Instruct	qwen-7b	gsm8k	CoT	1000	88.7	[86.6, 90.5]
Qwen2.5-Instruct	qwen-7b	gsm8k	structured	1000	22.4	[19.9, 25.1]
Qwen2.5-Instruct	qwen-7b	mmlu	zero-shot	1000	70.2	[67.3, 73.0]
Qwen2.5-Instruct	qwen-7b	mmlu	few-shot	1000	71.7	[68.8, 74.4]
Qwen2.5-Instruct	qwen-7b	mmlu	CoT	1000	54.7	[51.6, 57.8]
Qwen2.5-Instruct	qwen-7b	mmlu	structured	1000	69.9	[67.0, 72.7]
Llama-3	llama-1b	sst2	zero-shot	872	58.4	[55.1, 61.6]
Llama-3	llama-1b	sst2	few-shot	872	0.1	[0.0, 0.6]
Llama-3	llama-1b	sst2	CoT	872	70.6	[67.5, 73.6]
Llama-3	llama-1b	sst2	structured	872	73.4	[70.4, 76.2]
Llama-3	llama-1b	gsm8k	zero-shot	1000	6.6	[5.2, 8.3]
Llama-3	llama-1b	gsm8k	few-shot	1000	1.4	[0.8, 2.3]
Llama-3	llama-1b	gsm8k	CoT	1000	1.4	[0.8, 2.3]
Llama-3	llama-1b	gsm8k	structured	1000	4.8	[3.6, 6.3]
Llama-3	llama-1b	mmlu	zero-shot	1000	32.4	[29.6, 35.4]
Llama-3	llama-1b	mmlu	few-shot	1000	1.3	[0.8, 2.2]
Llama-3	llama-1b	mmlu	CoT	1000	33.3	[30.4, 36.3]
Llama-3	llama-1b	mmlu	structured	1000	39.0	[36.0, 42.1]
Llama-3	llama-3b	sst2	zero-shot	872	88.6	[86.4, 90.6]
Llama-3	llama-3b	sst2	few-shot	872	55.7	[52.4, 59.0]
Llama-3	llama-3b	sst2	CoT	872	84.7	[82.2, 87.0]
Llama-3	llama-3b	sst2	structured	872	89.9	[87.7, 91.7]
Llama-3	llama-3b	gsm8k	zero-shot	1000	11.1	[9.3, 13.2]
Llama-3	llama-3b	gsm8k	few-shot	1000	1.5	[0.9, 2.5]
Llama-3	llama-3b	gsm8k	CoT	1000	73.0	[70.2, 75.7]
Llama-3	llama-3b	gsm8k	structured	1000	7.7	[6.2, 9.5]
Llama-3	llama-3b	mmlu	zero-shot	1000	52.9	[49.8, 56.0]
Llama-3	llama-3b	mmlu	few-shot	1000	39.7	[36.7, 42.8]
Llama-3	llama-3b	mmlu	CoT	1000	55.9	[52.8, 58.9]
Llama-3	llama-3b	mmlu	structured	1000	56.6	[53.5, 59.6]

Llama-3	llama-8b	sst2	zero-shot	872	92.9	[91.0, 94.4]
Llama-3	llama-8b	sst2	few-shot	872	85.4	[82.9, 87.6]
Llama-3	llama-8b	sst2	CoT	872	90.9	[88.9, 92.7]
Llama-3	llama-8b	sst2	structured	872	92.3	[90.4, 93.9]
Llama-3	llama-8b	gsm8k	zero-shot	1000	20.1	[17.7, 22.7]
Llama-3	llama-8b	gsm8k	few-shot	1000	0.8	[0.4, 1.6]
Llama-3	llama-8b	gsm8k	CoT	1000	81.9	[79.4, 84.2]
Llama-3	llama-8b	gsm8k	structured	1000	9.9	[8.2, 11.9]
Llama-3	llama-8b	mmlu	zero-shot	1000	62.5	[59.5, 65.4]
Llama-3	llama-8b	mmlu	few-shot	1000	56.3	[53.2, 59.3]
Llama-3	llama-8b	mmlu	CoT	1000	57.3	[54.2, 60.3]
Llama-3	llama-8b	mmlu	structured	1000	63.6	[60.6, 66.5]
Gemma-2	gemma-2b	sst2	zero-shot	872	89.9	[87.7, 91.7]
Gemma-2	gemma-2b	sst2	few-shot	872	74.3	[71.3, 77.1]
Gemma-2	gemma-2b	sst2	CoT	872	90.5	[88.4, 92.3]
Gemma-2	gemma-2b	sst2	structured	872	88.6	[86.4, 90.6]
Gemma-2	gemma-2b	gsm8k	zero-shot	1000	5.7	[4.4, 7.3]
Gemma-2	gemma-2b	gsm8k	few-shot	1000	2.5	[1.7, 3.7]
Gemma-2	gemma-2b	gsm8k	CoT	1000	54.0	[50.9, 57.1]
Gemma-2	gemma-2b	gsm8k	structured	1000	5.4	[4.2, 7.0]
Gemma-2	gemma-2b	mmlu	zero-shot	1000	55.2	[52.1, 58.3]
Gemma-2	gemma-2b	mmlu	few-shot	1000	52.1	[49.0, 55.2]
Gemma-2	gemma-2b	mmlu	CoT	1000	39.1	[36.1, 42.2]
Gemma-2	gemma-2b	mmlu	structured	1000	56.5	[53.4, 59.5]
Gemma-2	gemma-9b	sst2	zero-shot	872	90.6	[88.5, 92.4]
Gemma-2	gemma-9b	sst2	few-shot	872	89.8	[87.6, 91.6]
Gemma-2	gemma-9b	sst2	CoT	872	90.6	[88.5, 92.4]
Gemma-2	gemma-9b	sst2	structured	872	92.2	[90.2, 93.8]
Gemma-2	gemma-9b	gsm8k	zero-shot	1000	16.3	[14.1, 18.7]
Gemma-2	gemma-9b	gsm8k	few-shot	1000	2.7	[1.9, 3.9]
Gemma-2	gemma-9b	gsm8k	CoT	1000	86.3	[84.0, 88.3]
Gemma-2	gemma-9b	gsm8k	structured	1000	21.8	[19.4, 24.5]
Gemma-2	gemma-9b	mmlu	zero-shot	1000	69.7	[66.8, 72.5]
Gemma-2	gemma-9b	mmlu	few-shot	1000	69.1	[66.2, 71.9]
Gemma-2	gemma-9b	mmlu	CoT	1000	70.3	[67.4, 73.1]
Gemma-2	gemma-9b	mmlu	structured	1000	68.3	[65.4, 71.1]

C.2 McNemar tests vs. zero-shot (Holm-adjusted)

Family	Model	Task	Comparison	p (adj)	
Qwen2.5-Instruct	qwen-0.5b	sst2	zero-shot vs few-shot	0.0000	***
Qwen2.5-Instruct	qwen-0.5b	sst2	zero-shot vs CoT	0.0002	***
Qwen2.5-Instruct	qwen-0.5b	sst2	zero-shot vs structured	0.0000	***
Qwen2.5-Instruct	qwen-0.5b	gsm8k	zero-shot vs few-shot	0.6733	ns
Qwen2.5-Instruct	qwen-0.5b	gsm8k	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-0.5b	gsm8k	zero-shot vs structured	0.6733	ns
Qwen2.5-Instruct	qwen-0.5b	mmlu	zero-shot vs few-shot	0.0000	***
Qwen2.5-Instruct	qwen-0.5b	mmlu	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-0.5b	mmlu	zero-shot vs structured	0.0000	***
Qwen2.5-Instruct	qwen-1.5b	sst2	zero-shot vs few-shot	0.0000	***
Qwen2.5-Instruct	qwen-1.5b	sst2	zero-shot vs CoT	0.0040	**
Qwen2.5-Instruct	qwen-1.5b	sst2	zero-shot vs structured	0.0000	***
Qwen2.5-Instruct	qwen-1.5b	gsm8k	zero-shot vs few-shot	0.0001	***
Qwen2.5-Instruct	qwen-1.5b	gsm8k	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-1.5b	gsm8k	zero-shot vs structured	0.0000	***
Qwen2.5-Instruct	qwen-1.5b	mmlu	zero-shot vs few-shot	0.5345	ns
Qwen2.5-Instruct	qwen-1.5b	mmlu	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-1.5b	mmlu	zero-shot vs structured	0.5345	ns
Qwen2.5-Instruct	qwen-3b	sst2	zero-shot vs few-shot	0.1987	ns
Qwen2.5-Instruct	qwen-3b	sst2	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-3b	sst2	zero-shot vs structured	0.0000	***
Qwen2.5-Instruct	qwen-3b	gsm8k	zero-shot vs few-shot	1.0000	ns
Qwen2.5-Instruct	qwen-3b	gsm8k	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-3b	gsm8k	zero-shot vs structured	1.0000	ns
Qwen2.5-Instruct	qwen-3b	mmlu	zero-shot vs few-shot	0.7927	ns
Qwen2.5-Instruct	qwen-3b	mmlu	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-3b	mmlu	zero-shot vs structured	0.9267	ns
Qwen2.5-Instruct	qwen-7b	sst2	zero-shot vs few-shot	0.0000	***
Qwen2.5-Instruct	qwen-7b	sst2	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-7b	sst2	zero-shot vs structured	0.0000	***
Qwen2.5-Instruct	qwen-7b	gsm8k	zero-shot vs few-shot	0.0000	***
Qwen2.5-Instruct	qwen-7b	gsm8k	zero-shot vs CoT	0.0000	***
Qwen2.5-Instruct	qwen-7b	gsm8k	zero-shot vs structured	0.0000	***
Qwen2.5-Instruct	qwen-7b	mmlu	zero-shot vs few-shot	0.2396	ns
Qwen2.5-Instruct	qwen-7b	mmlu	zero-shot vs CoT	0.0000	***

Qwen2.5-Instruct	qwen-7b	mmlu	zero-shot vs structured	0.8174	ns
Llama-3	llama-1b	sst2	zero-shot vs few-shot	0.0000	***
Llama-3	llama-1b	sst2	zero-shot vs CoT	0.0000	***
Llama-3	llama-1b	sst2	zero-shot vs structured	0.0000	***
Llama-3	llama-1b	gsm8k	zero-shot vs few-shot	0.0000	***
Llama-3	llama-1b	gsm8k	zero-shot vs CoT	0.0000	***
Llama-3	llama-1b	gsm8k	zero-shot vs structured	0.0923	ns
Llama-3	llama-1b	mmlu	zero-shot vs few-shot	0.0000	***
Llama-3	llama-1b	mmlu	zero-shot vs CoT	0.6746	ns
Llama-3	llama-1b	mmlu	zero-shot vs structured	0.0018	**
Llama-3	llama-3b	sst2	zero-shot vs few-shot	0.0000	***
Llama-3	llama-3b	sst2	zero-shot vs CoT	0.0030	**
Llama-3	llama-3b	sst2	zero-shot vs structured	0.1093	ns
Llama-3	llama-3b	gsm8k	zero-shot vs few-shot	0.0000	***
Llama-3	llama-3b	gsm8k	zero-shot vs CoT	0.0000	***
Llama-3	llama-3b	gsm8k	zero-shot vs structured	0.0053	**
Llama-3	llama-3b	mmlu	zero-shot vs few-shot	0.0000	***
Llama-3	llama-3b	mmlu	zero-shot vs CoT	0.0853	ns
Llama-3	llama-3b	mmlu	zero-shot vs structured	0.0118	*
Llama-3	llama-8b	sst2	zero-shot vs few-shot	0.0000	***
Llama-3	llama-8b	sst2	zero-shot vs CoT	0.1082	ns
Llama-3	llama-8b	sst2	zero-shot vs structured	0.5596	ns
Llama-3	llama-8b	gsm8k	zero-shot vs few-shot	0.0000	***
Llama-3	llama-8b	gsm8k	zero-shot vs CoT	0.0000	***
Llama-3	llama-8b	gsm8k	zero-shot vs structured	0.0000	***
Llama-3	llama-8b	mmlu	zero-shot vs few-shot	0.0000	***
Llama-3	llama-8b	mmlu	zero-shot vs CoT	0.0065	**
Llama-3	llama-8b	mmlu	zero-shot vs structured	0.3786	ns
Gemma-2	gemma-2b	sst2	zero-shot vs few-shot	0.0000	***
Gemma-2	gemma-2b	sst2	zero-shot vs CoT	0.6301	ns
Gemma-2	gemma-2b	sst2	zero-shot vs structured	0.2367	ns
Gemma-2	gemma-2b	gsm8k	zero-shot vs few-shot	0.0009	***
Gemma-2	gemma-2b	gsm8k	zero-shot vs CoT	0.0000	***
Gemma-2	gemma-2b	gsm8k	zero-shot vs structured	0.8011	ns
Gemma-2	gemma-2b	mmlu	zero-shot vs few-shot	0.0467	*
Gemma-2	gemma-2b	mmlu	zero-shot vs CoT	0.0000	***
Gemma-2	gemma-2b	mmlu	zero-shot vs structured	0.2183	ns

Gemma-2	gemma-9b	sst2	zero-shot vs few-shot	0.9651	ns
Gemma-2	gemma-9b	sst2	zero-shot vs CoT	0.9651	ns
Gemma-2	gemma-9b	sst2	zero-shot vs structured	0.2143	ns
Gemma-2	gemma-9b	gsm8k	zero-shot vs few-shot	0.0000	***
Gemma-2	gemma-9b	gsm8k	zero-shot vs CoT	0.0000	***
Gemma-2	gemma-9b	gsm8k	zero-shot vs structured	0.0000	***
Gemma-2	gemma-9b	mmlu	zero-shot vs few-shot	1.0000	ns
Gemma-2	gemma-9b	mmlu	zero-shot vs CoT	1.0000	ns
Gemma-2	gemma-9b	mmlu	zero-shot vs structured	0.4682	ns

Significance: *** p<.001, ** p<.01, * p<.05, ns = not significant.

C.3 Full-precision (bf16) control: Qwen ladder (§4.4)

The Qwen 0.5B / 1.5B / 3B checkpoints were re-run at full precision (bf16) on the same fixed 1000-item MMLU and GSM8K samples used for the 4-bit results. On MMLU the CoT penalty reproduces and grows with scale (−12.9 / −14.2 / −20.9pp, against −7.2 / −10.3 / −13.7pp at 4-bit); on GSM8K the exchange rate holds, with CoT far above zero-shot at every size. Every zero-shot-vs-CoT McNemar comparison below is significant at $p < .001$.

Family	Model	Task	Strategy	n	Acc (%)	95% CI
Qwen2.5 (bf16)	qwen-0.5b	mmlu	zero-shot	1000	43.1	[40.1, 46.2]
Qwen2.5 (bf16)	qwen-0.5b	mmlu	CoT	1000	30.2	[27.4, 33.1]
Qwen2.5 (bf16)	qwen-0.5b	gsm8k	zero-shot	1000	2.8	[1.9, 4.0]
Qwen2.5 (bf16)	qwen-0.5b	gsm8k	CoT	1000	43.1	[40.1, 46.2]
Qwen2.5 (bf16)	qwen-1.5b	mmlu	zero-shot	1000	57.8	[54.7, 60.8]
Qwen2.5 (bf16)	qwen-1.5b	mmlu	CoT	1000	43.6	[40.6, 46.7]
Qwen2.5 (bf16)	qwen-1.5b	gsm8k	zero-shot	1000	5.8	[4.5, 7.4]
Qwen2.5 (bf16)	qwen-1.5b	gsm8k	CoT	1000	66.0	[63.0, 68.9]
Qwen2.5 (bf16)	qwen-3b	mmlu	zero-shot	1000	64.0	[61.0, 66.9]
Qwen2.5 (bf16)	qwen-3b	mmlu	CoT	1000	43.1	[40.1, 46.2]
Qwen2.5 (bf16)	qwen-3b	gsm8k	zero-shot	1000	1.6	[1.0, 2.6]
Qwen2.5 (bf16)	qwen-3b	gsm8k	CoT	1000	83.2	[80.8, 85.4]

McNemar (zero-shot vs CoT), Holm-adjusted within model×task: all six comparisons *** ($p < .001$). MMLU deltas −12.9 / −14.2 / −20.9pp; GSM8K deltas +40.3 / +60.2 / +81.6pp.

Appendix D. Compute and reproducibility

All experiments ran on a single Apple M4 laptop (16 GB unified memory) via MLX, one model resident at a time. Models are the 4-bit mlx-community quantizations of Qwen2.5-Instruct (0.5/1.5/3/7B), Llama-3 (Llama-3.2-1B/3B, Meta-Llama-3.1-8B), and Gemma-2 (2B, 9B). Decoding is greedy (temperature 0). Max new tokens per condition: 8 (MMLU), 16 (SST-2), 48 (GSM8K non-

CoT), 256 (SST-2/MMLU CoT), 512 (GSM8K CoT). Each condition is scored on the same fixed sample, held constant across all models and strategies: the full SST-2 validation set (872 items) and a fixed sample of 1000 items each for GSM8K and MMLU, with MMLU stratified across all 57 subjects. The full harness (`run_experiments_local.py`), prompt builder, extraction code, prompt templates, and per-example results are released at github.com/ssamalsamir/cot-model-specificity. Running `--aggregate` reproduces all tables from the saved per-example CSVs.